

Enhancing Accessibility on Android



Life is not easy for the physically challenged. Thankfully, technology is helping those who are physically challenged to enjoy and experience Web and phone interfaces just like normal people. Accessibility options on Android help to make the OS more inclusive. Here are a few valuable suggestions for mobile app developers on how to enhance accessibility for the physically challenged.

According to a report by the World Health Organisation (WHO), the number of visually impaired people in the world is 285 million! The report also states that 90 per cent of these people live in low-income settings. Android, being an open source platform, has been adopted by various handset manufacturers. Owing to the open source nature of Android, and the availability of many low-cost Android powered smartphones in the market, the operating system has become extremely popular in developing countries. Hence, it is of utmost importance that Android applications are easy to use for people with and without impairments. Users should not feel that the application is not meant to be used by them because of their impairment, i.e., they should not feel disabled. Hence, stakeholders, designers and developers need to take into account the opinions of users with impairments and the problems they face while using an application, in order to make it accessible to everyone.

Principles of accessibility

The World Wide Web Consortium (W3C) has defined the following four principles of accessibility, based on which the Web Content Accessibility Guidelines (WCAG) were developed. These should be followed by those developing

web applications, and can be applied to the development of mobile applications as well.

1. **Perceivable:** The information being displayed should be perceivable, i.e., text alternatives should be provided for images; audio and video content should be accompanied by captions, while colours should not confuse the user.
2. **Operable:** It must be easy for users to navigate across multiple screens, whether they are using a touchscreen device or a keyboard. The components on the screen should be easy to operate.
3. **Understandable:** The content presented to the user should be easy to read and understand.
4. **Robust:** The application should be compatible with maximum number of assistive technologies, web browsers, etc.

Accessibility options for Android users

Through Lollipop, Google offers a set of accessibility options that the user can control from a single screen. This article explains the options available in Android 5.0, which can be used by people with visual, motor or auditory impairment (Figure 1).

1. **TalkBack:** Talkback is an accessibility service that consists of many features, such as 'Explore by Touch' and 'Screen Reader'. It is meant to be used by people who have visual